ARC-CONNECTIVITY IN DIRECTED MULTIGRAPHS

Abstract. For a loopless directed multigraph, we determine the maximum number of arcs when every local arc-connectivity is at most a prescribed value. The proof combines a sharp bound for reachability-preserving spanning subdigraphs with an induction that removes one such subdigraph at a time.

For a loopless directed multigraph D, let $A(D)$ denote its arc multiset, and let $\lambda_D(u,v)$ be the maximum number of pairwise arc-disjoint directed u–v paths. Set

$$\lambda^{\max}(D) = \max_{u\neq v} \lambda_D(u,v), \qquad L_m^{\mathrm{dir}}(n) = \max\bigl\{|A(D)| : |V(D)| = n,\ \lambda^{\max}(D) \le m-1\bigr\},$$

where arcs are counted with multiplicity, and write

$$M(n) = \max\left\{2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right\}.$$

Theorem 1. *For all $n \ge 2$ and $m \ge 2$,*

$$L_m^{\mathrm{dir}}(n) = (m-1)M(n).$$

A spanning subdigraph $R \subseteq D$ is *reachability-preserving* if u reaches v in R exactly when u reaches v in D.

Lemma 2. *Every loopless directed multigraph D on $n \ge 2$ vertices has a reachability-preserving spanning subdigraph with at most $M(n)$ arcs.*

Proof. Let the strongly connected components of D have orders $n_1, \dots, n_q$. In each component, the union of an in-arborescence and an out-arborescence rooted at the same vertex is strongly connected and uses at most $2(n_i - 1)$ arcs.

Contract the components and suppress parallel arcs, obtaining a DAG Q_0 on q vertices. Choose an inclusion-minimal spanning subdigraph $Q \subseteq Q_0$ with the same reachability relation. The underlying graph of Q is triangle-free: an acyclic orientation of a triangle is necessarily

$$a \to b, \qquad b \to c, \qquad a \to c,$$

and then $a \to c$ is redundant. Hence Mantel's theorem gives $|A(Q)| \le \lfloor q^2/4 \rfloor$. For completeness, if a triangle-free graph H has q vertices and e edges, then

$$\sum_v d(v)^2 = \sum_{xy\in E(H)} (d(x)+d(y)) \le qe, \qquad \sum_v d(v)^2 \ge \frac{(2e)^2}{q},$$

so $e \le q^2/4$.

Lift one arc of D for each arc of Q and add the chosen arborescences. The resulting spanning subdigraph preserves reachability and has at most

$$2\sum_{i=1}^{q}(n_i - 1) + \left\lfloor \frac{q^2}{4} \right\rfloor = 2(n-q) + \left\lfloor \frac{q^2}{4} \right\rfloor$$

arcs. If $f(q)$ denotes the last expression, then

$$f(q+1) - f(q) = -2 + \left\lfloor \frac{q+1}{2} \right\rfloor$$

is nondecreasing. Thus f is discretely convex, and its maximum on $1 \le q \le n$ is attained at an endpoint. Since $f(1) = 2(n-1)$ and $f(n) = \lfloor n^2/4 \rfloor$, we have $f(q) \le M(n)$. □

Lemma 3. *Let D satisfy $\lambda^{\max}(D) \le k$, where $k \ge 1$, and let $R \subseteq D$ be reachability-preserving. Write $D - R$ for the multigraph obtained by deleting the particular arc copies of R. Then*

$$\lambda^{\max}(D - R) \le k - 1.$$

Proof. Put $D' = D - R$. If D' contains $t \geq 1$ arc-disjoint u–v paths, then R contains a directed u–v path. This additional path is arc-disjoint from the first t, so $t + 1 \leq \lambda_D(u, v) \leq k$. Hence $t \leq k - 1$ for every ordered pair (u, v). □

Proof of Theorem 1. Put $k = m - 1$. We prove by induction on k that every n-vertex directed multigraph D with $\lambda^{\max}(D) \leq k$ has at most $kM(n)$ arcs. The assertion is immediate for $k = 0$. For $k \geq 1$, choose R as in Lemma 2. By Lemma 3, the residual multigraph $D - R$ has maximum local arc-connectivity at most $k - 1$. Therefore

$$|A(D)| = |A(R)| + |A(D - R)| \leq M(n) + (k - 1)M(n) = kM(n).$$

For equality on the linear branch, replace every edge of a spanning tree by k parallel arcs in each direction. This gives $2k(n - 1)$ arcs and local arc-connectivity exactly k: the unique underlying tree path supplies k arc-disjoint directed paths, while the k arcs in the appropriate direction across any edge of that path form a cut.

For equality on the quadratic branch, partition the vertices into sets X, Y of orders $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$, and place k parallel arcs from every vertex of X to every vertex of Y. This gives $k\lfloor n^2/4 \rfloor$ arcs and maximum local arc-connectivity k. Taking the better construction proves the result. □

Corollary 4.

$$L_3^{\mathrm{dir}}(7) = 24.$$

Proof. This is Theorem 1 with $m = 3$ and $n = 7$. □